

Tushar R C Raja

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Education

SPIT Mumbai, B.Tech in Computer Engineering

Nov 2022 - Present

- CGPA: 9.84/10

- **Major Coursework:** Data Structures and Algorithms, Data Analysis, Machine Learning

DPS Moradabad, CBSE Board

2022

- Class XII: 98%
- Class X: 98%

Projects

OrgDaan: Liver Health Assessment and Matching Compatibility Algorithm

- Engineered a 'risk score' synthetic variable by incorporating hyperparameter tuning and a hybrid-classifier pipeline combining Random Forest and Logistic Regression giving an accuracy of 82%.
- **Tech Stack:** Machine Learning, Python, Jupyter

PantryPal: Grocery Ordering and Management Site

GitHub

- Built a full-stack grocery e-commerce site with demand forecasting, dynamic pricing, and personalized recommendations.
- Integrated an NLP chatbot for voice-based shopping and implemented secure payment and authentication systems.
- **Tech Stack:** MERN, Braintree, JWT, OAuth2

FilmInsight: A Machine Learning-Based Movie Recommendation System

GitHub

- Developed a hybrid movie recommender leveraging NLP-based plot embeddings and metadata analysis (genre, cast, director).
- Integrated the IMDb API for real-time data and deployed the system using Streamlit to create an interactive user experience.
- **Tech Stack:** Python, Scikit-learn, NLP, IMDb API, Streamlit, Pandas, NumPy

Alertify: A Disaster Predictor

GitHub

- Developed an intelligent model that analyzes situational data to predict the probability of various disasters, including natural calamities, war, and shifts in global trade relations.
- **Tech Stack:** Python, NLP (spaCy/BERT), Anomaly Detection, Pandas, Streamlit, Scikit-learn, Hyperparameter Tuning, QGIS

BowlGuard: Bio-Mechanical Analysis of Bowling Actions

- Predicted a bowler's injury risk level by conducting a biomechanical analysis of their bowling action.
- **Tech Stack:** Python, YOLO, Machine Learning, GridSearch, Hyperparameter Tuning, OpenCV, Haar Wavelet Analysis

Skills

Languages & Libraries: C++ (Primary), Java, SQL, HTML, CSS, MongoDB, Express, Node.js, Matplotlib, Pandas, NLTK

Technical Skills: Database Management, Software Engineering, Operating Systems, Artificial Intelligence, Machine Learning, Data Analytics

Extracurricular Activities & Achievements

- **LeetCode:** Knight (1900+ Rating), solved over 650 DSA problems.
- **Codeforces:** Pupil (1328 Rating).
- **Flipkart Grid 6.0:** Qualified for Level 2 of the hackathon, competing against over 50,000 participants.
- **AIravat 2.0 Hackathon:** Achieved Top 3 in the Machine Learning domain at SPIT Mumbai.
- **Club of Coders (CoC):** Awarded premium membership after securing Rank 3 in a competitive programming round among 150 participants.
- **Student Representative:** Selected for Open Elective (OE) courses in NLP and Data Analytics across multiple branches.
- **Research Publications (In Progress):** Authored two papers on land cover change in Navi Mumbai and findings from the BowlGuard project.
- **Olympiads:** Gold Medalist, IMO (Regionals 2018); Silver Medalist, NCO (District 2017).
- **HSS Psychology:** Completed Levels I, II, and III, ranking in the top 1st percentile.